B. Tech – I Semester
End Semester Exam

Time Allowed: 3 Hours
Roll No. ______________
Maximum Marks: 50

Note: ***Section C is compulsory. Attempt any six questions by selecting three questions from Section A & three questions from Section B.***

Section-A

1. (a) Using Cayley Hamilton theorem find the inverse of the matrix $\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$. 4

   (b) If λ is an eigen value of a matrix A, show that λ^3 is an eigen value of A^3. 1

2. Diagonalize the matrix $A=\begin{pmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ and hence find A^8. 5

3. (a) Reduce the matrix $\begin{pmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{pmatrix}$ to the normal form, hence find its rank. . 3

   (b) For what values of a and b do the system of equations

   $$x+y+z=6,\ x+2y+3z=10,\ x+2y+az=b$$

   have *(i)* No solution *(ii)* A unique solution. 2

4. Find the dimensions of the rectangular box, open at the top, of maximum capacity whose surface is 432 square cm. 5

5. If $u=\operatorname{cosec}^{-1}\left(\dfrac{x^{\frac{1}{2}}+y^{\frac{1}{2}}}{x^{\frac{1}{3}}+y^{\frac{1}{3}}}\right)^{\frac{1}{2}}$, show that $x^2\dfrac{\partial^2 u}{\partial x^2}+2xy\dfrac{\partial^2 u}{\partial x\partial y}+y^2\dfrac{\partial^2 u}{\partial y^2}=\dfrac{\tan u}{144}\left(13+\tan^2 u\right)$. 5

Section-B

6. (a) Find an equation whose roots are the nth powers of the roots of the equation $x^2-2x\cos\theta+1=0$. 2

   (b) Show that the function $f(z)=\sqrt{|xy|}$ is not analytic at the origin even though Cauchy Riemann equations are satisfied thereat. 3

7. If $f(z)=u+iv$ is an analytic function, find $f(z)$ if $u-v=e^x(\cos y-\sin y)$. 5

8. (a) Prove that $\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$ 2

(b) Discuss the convergence of the integral $\int_{-\infty}^{\infty} \frac{1}{x^2+2x+2}\, dx$. 3

9. Prove that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. 5

10. (a) Show that $\int_0^{\infty}\left(\frac{1}{x+1} - e^{-x}\right)\frac{dx}{x}$ is convergent. 3

(b) Evaluate $\int_0^{\infty} \frac{x^4(1-x^5)}{(1+x)^{15}}.dx$. 2

Section-C

11. (a) Show that eigen values of a Skew Hermitian matrix are either zero or purely imaginary.

(b) Prove that the matrix $\begin{pmatrix} -\frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$ is orthogonal.

(c) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$

(d) If $z = f(x+ay) + \phi(x-ay)$ prove that $\frac{\partial^2 z}{\partial y^2} = a^2\frac{\partial^2 z}{\partial x^2}$.

(e) Prove that $\frac{1}{2}\begin{pmatrix} 1+i & -1+i \\ 1+i & 1-i \end{pmatrix}$ is a unitary matrix.

(f) Show that real and imaginary parts of an analytic function are harmonic functions.

(g) Prove that $\Gamma(n) = (n-1)!$, where n is a positive integer.

(h) Determine a, b, c, d so that the function $f(z) = (x^2 + axy + by^2) + i(cx^2 + dxy + y^2)$ is analytic.

(i) Show that $32\cos^6\theta = \cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10$.

(j) Using Beta and Gamma functions, prove that $\int_0^{\frac{\pi}{2}} \sin^2\theta\cos^4\theta.d\theta = \frac{\pi}{32}$